Predicates and Quantifiers

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Predicate Logic

Why Predicate Logic is Needed?

- Propositional logic can only handle whole statements (True/False).
- It cannot express statements involving variables.
- Example: Propositional logic cannot directly express –
  - *"Every computer connected to the university network is functioning properly."*
  - *"There is a computer that is under attack by an intruder."*
- Predicate Logic allows us to capture such statements using variables and quantifiers.

Definition and Representation

Predicate

A predicate is a statement involving variables that becomes a proposition once specific values are assigned.

Examples:

- $P(x) : x > 3$
- $Q(x, y) : x = y + 3$
- $R(x, y, z) : x + y = z$
- $A(x)$: "Computer x is under attack by an intruder"

Example Problem 1

Let $P(x) : x > 3$. Find the truth values of $P(4)$ and $P(2)$.

Solution

- $P(4) : 4 > 3$ is True.
- $P(2) : 2 > 3$ is False.

Example Problem 2

Let $Q(x, y) : x = y + 3$. Find the truth values of $Q(1, 2)$ and $Q(3, 0)$.

Solution

- $Q(1, 2) : 1 = 2 + 3$ is False.
- $Q(3, 0) : 3 = 0 + 3$ is True.

Example Problem 3

Let $R(x, y, z) : x + y = z$. Find the truth values of $R(1, 2, 3)$ and $R(0, 0, 1)$.

Solution

- $R(1, 2, 3) : 1 + 2 = 3$ is True.
- $R(0, 0, 1) : 0 + 0 = 1$ is False.

Example Problem 4

Let $A(x)$: "Computer x is under attack by an intruder."
Assume CS2 and MATH1 are under attack. Find the truth values of $A(CS1)$, $A(CS2)$, $A(MATH1)$.

Solution

- $A(CS1)$: False (CS1 not under attack).
- $A(CS2)$: True (CS2 is under attack).
- $A(MATH1)$: True (MATH1 is under attack).

Quantifiers

Quantifiers

- Quantification expresses the extent to which a predicate is true over a range of elements.
- Two main types:
  - Universal Quantifier ($\forall$): "for all"
  - Existential Quantifier ($\exists$): "there exists"
- Words like *all*, *some*, *many*, *none*, *few* are natural language indicators of quantifiers.
- The area of logic that deals with predicates and quantifiers is called **predicate calculus**.

Domain of Discourse

- The domain of discourse (or simply, domain) specifies the set of values that variables can take.
- Example domains:
  - All real numbers.
  - Positive integers not exceeding 4.
  - All computers on a university network.
- The truth value of a quantified statement depends on the chosen domain.

Universal Quantifier

Definition

The universal quantification of $P(x)$ is the statement:

$$\forall x\, P(x) : "P(x) \text{ is true for all values of } x \text{ in the domain."}$$

- True if $P(x)$ is true for every element in the domain.
- False if there exists at least one counterexample.

Existential Quantifier

Definition

The existential quantification of $P(x)$ is the statement:

$\exists x\, P(x)$: "There exists an element x in the domain such that $P(x)$ is true."

- True if $P(x)$ is true for at least one element in the domain.
- False if $P(x)$ is false for all elements.

Truth Conditions for Quantifiers

Statement	When True?	When False?
$\forall x\, P(x)$	$P(x)$ true for every x	$P(x)$ false for some x
$\exists x\, P(x)$	$P(x)$ true for some x	$P(x)$ false for every x

Example Problem 1

Let $P(x) : x + 1 > x$, domain = all real numbers. Find truth value of $\forall x P(x)$.

Solution

- For all real x, $x + 1 > x$ is true.
- Hence, $\forall x P(x)$ is true.

Example Problem 2

Let $Q(x) : x < 2$, domain = all real numbers. Find truth value of $\forall x Q(x)$.

Solution

- $Q(3) : 3 < 2$ is false, so not all x satisfy $Q(x)$.
- Therefore, $\forall x Q(x)$ is false.

Example Problem 3

Let $P(x) : x^2 \geq x$. Find truth value of $\forall x P(x)$ with domain $=$

- All real numbers.
- All integers.

Solution

- For reals: False (counterexample $x = 1/2$).
- For integers: True (no integer between 0 and 1).

Example Problem 4

Let $Q(x) : x = x + 1$, domain = all real numbers. Find truth value of $\exists x Q(x)$.

Solution

- $x = x + 1$ is never true for any real x.
- Thus, $\exists x Q(x)$ is false.

Example Problem 5

Let $P(x) : x^2 > 10$, domain $= \{1, 2, 3, 4\}$. Find truth value of $\exists x P(x)$.

Solution

- $P(4) : 4^2 > 10$ is true.
- Therefore, $\exists x P(x)$ is true.

Negating Quantifiers

- Negation of quantifiers follows rules similar to De Morgan's Laws.
- The quantifier changes when negated:
  - Negation of a universal statement ($\forall$) becomes an existential statement ($\exists$).
  - Negation of an existential statement ($\exists$) becomes a universal statement ($\forall$).

De Morgan's Laws for Quantifiers

Negation	Equivalent	When True?	When False?
$\neg\exists x P(x)$	$\forall x \neg P(x)$	$P(x)$ false for every x	$P(x)$ true for some x
$\neg\forall x P(x)$	$\exists x \neg P(x)$	$P(x)$ false for some x	$P(x)$ true for every x

Example Problem 1

What is the negation of $\forall x(x^2 > x)$?

Solution

- Negation: $\neg\forall x(x^2 > x)$
- Equivalent to: $\exists x\neg(x^2 > x)$
- Simplifies to: $\exists x(x^2 \leq x)$

Example Problem 2

What is the negation of $\exists x(x^2 = 2)$?

Solution

- Negation: $\neg\exists x(x^2 = 2)$
- Equivalent to: $\forall x\neg(x^2 = 2)$
- Simplifies to: $\forall x(x^2 \neq 2)$

Nested Quantifiers

Definition of Nested Quantifiers

- A **nested quantifier** occurs when one quantifier is within the scope of another.
- Example: $\forall x \exists y (x + y = 0)$
- These allow us to express more complex mathematical statements.
- The order of quantifiers is important and can change the meaning of the statement.

Examples of Nested Quantifiers

Domain: All Real Numbers

- $\forall x \forall y(x + y = y + x)$ (Commutative law of addition)
- $\forall x \exists y(x + y = 0)$ (Additive inverse property)
- $\forall x \forall y \forall z(x + (y + z) = (x + y) + z)$ (Associative law of addition)

Translating Nested Quantifiers into English

Example: $\forall x \forall y(((x > 0) \wedge (y < 0)) \rightarrow (xy < 0))$, domain = all real numbers.

Translation

For every real number x and for every real number y, if $x > 0$ and $y < 0$, then $xy < 0$.

Simplified English: *The product of a positive real number and a negative real number is always negative.*

Example Problem 1

Let $P(x, y) : x + y = y + x$, domain = all real numbers. What is the truth value of $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$?

Solution

- Both express commutativity of addition.
- Since $x + y = y + x$ is true for all reals, both quantifications are true.

Example Problem 2

Domain

$$A = \{-2, 0, 1, 3\}, \qquad x, y \in A$$

Statement (over $x, y \in A$)	**Truth**
$\forall x \, \exists y \, (x + y \leq 1)$	?
$\forall x \, \forall y \, (x + y \leq 1)$	?
$\forall x \, \forall y \, (x + y \leq 10)$	?
$\exists x \, \forall y \, (xy = 0)$	?
$\exists x \, \forall y \, (x + y = 2)$	?
$\exists x \, \exists y \, (x + y = -1)$	?

Example Problem 2

Domain

$$A = \{-2, 0, 1, 3\}, \qquad x, y \in A$$

Statement (over $x, y \in A$)	**Truth**
$\forall x \, \exists y \, (x + y \leq 1)$	True
$\forall x \, \forall y \, (x + y \leq 1)$	False
$\forall x \, \forall y \, (x + y \leq 10)$	True
$\exists x \, \forall y \, (xy = 0)$	True
$\exists x \, \forall y \, (x + y = 2)$	False
$\exists x \, \exists y \, (x + y = -1)$	True

Example Problem 3

Let $Q(x, y, z) : x + y = z$, domain = all real numbers. Evaluate $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$.

Solution

- $\forall x \forall y \exists z (x + y = z)$: True, since for any x, y, we can pick $z = x + y$.
- $\exists z \forall x \forall y (x + y = z)$: False, since no single z works for all x, y.